# Literature

# Literature Review — VLM Hallucination & Attention Intervention

> Focus: training-free inference-time methods to reduce hallucination / improve visual grounding in MLLMs.
> Format per paper: **Idea · Problem · Solution · Details · Datasets · GAPS**

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## Summary Table

| Paper | arXiv | Venue | Arch Tested | Core Idea |
|---|---|---|---|---|
| [AdaptVis](#adaptvis)<br><br>Why Is Spatial Reasoning Hard for VLMs? An Attention Mechanism Perspective on Focus Areas | 2503.01773 | ICML 2025 | LLaVA-1.5/1.6<br><br>WhatsUp, VSR | Use generation confidence to decide whether to **sharpen** (focus) or **broaden** image-token attention —<br>high confidence = sharpen, low = spread (Temperature) |
| ClearSight<br><br><br>ClearSight: Visual Signal Enhancement for Object Hallucination Mitigation in MLLMs | 2503.13107 | CVPR 2025 | LLaVA-1.5 7B/13B, Qwen-VL-7B<br><br><br>POPE, MME, NoCaps | In middle transformer layers (8–15), **boost attention to image tokens** and simultaneously **suppress attention to system-prompt tokens**, applied only to top-50% vision-aware heads |
| VHR<br><br>Cracking the Code of Hallucination in LVLMs with Vision-aware Head Divergence | 2412.13949 | ACL 2025 | LLaVA-1.5-7B, InstructBLIP-7B<br><br><br>CHAIR, | Measure which attention heads **change most** when image is removed → those are "vision-aware"; scale up only their outputs during generation |

| Paper | arXiv | Venue | Arch Tested | Core Idea |
|---|---|---|---|---|
| | | | POPE,<br>LLaVA-Bench | |
| VISTA<br><br>The Hidden Life of Tokens: Reducing Hallucination of LVLMs via Visual Information Steering | 2502.03628 | arXiv | LLaVA-1.5, MiniGPT-4, InstructBLIP<br><br>CHAIR, POPE, MMHal, MME | Inject a **"visual direction"** into the residual stream each step + blend final logits with earlier layers where visual signal is still strong |
| VAR / VisAttnSink<br><br>See What You Are Told: Visual Attention Sink in Large Multimodal Models | 2503.03321 | arXiv | LLaVA-1.5/1.6, VILA, Qwen2-VL, InternVL2<br><br>VQAv2, MMVP, CHAIR, POPE | Certain visual tokens always attract high attention regardless of content ("sinks"); **steal** that wasted attention and give it to semantically relevant visual tokens |
| Localization Heads<br><br>Your Large Vision-Language Model Only Needs A Few Attention Heads For Visual Grounding | 2503.06287 | arXiv | LLaVA-1.5 (1.3B–13B)<br><br>RefCOCO, ReasonSeg | Frozen VLMs have **~3 specialized heads** that naturally localize objects; extract their attention maps → bounding boxes, no training needed |
| LLMind / BASS<br><br>LLMind: Bio-inspired Training-free Adaptive Visual Representations for VLMs | 2603.14882 | CVPR 26 | Qwen2.5-VL-4B, SmolVLM-2B, LLaVA-OV, Qwen3-VL-2B<br><br>VQAv2, Seed-Bench, A-OKVQA | Like human foveal vision: **warp the image** (Möbius transform) to zoom into task-relevant regions and compress background; optimize warp via gradient-free feedback from model |
| AIR<br><br>Look Carefully: Adaptive Visual | 2602.24041 | ICLR 2026 | LLaVA-1.5-7B, Qwen-VL-Chat, GLM-4V-9B | Compress visual tokens to a **prototype-based top-Q subset**; use **Optimal Transport** (Sinkhorn-Knopp) |

| Paper | arXiv | Venue | Arch Tested | Core Idea |
|---|---|---|---|---|
| Reinforcements in Multimodal Large Language Models for Hallucination Mitigation | | | CHAIR, POPE, MME, MMBench | to score hidden-state ↔ patch alignment in FFN layers; re-inject only best-aligned patches — token selectivity without any query/text signal |
| Saliency-R1<br><br>Saliency-R1: Enforcing Interpretable and Faithful Vision-language Reasoning via Saliency-map Alignment Reward | 2604.04500 | CVPR 2026 | Qwen2.5-VL-3B/7B<br><br>COCO Caption, OpenPSG, GranDf | Decompose token logits → **saliency maps** of visual patch contributions; use saliency ↔ GT bounding-box IoU as **GRPO reward** to fine-tune model to attend to correct regions (⚠️ **requires LoRA + GRPO training — not training-free**) |
| SPIN<br><br>Mitigating Hallucinations in VLMs through Image-Guided Head Suppression | 2505.16411 | arXiv May 2025 | LLaVA-1.5-7B<br><br>POPE, CHAIR, MME, MMHal | **Suppress** image-*inattentive* heads (inverse of VHR): for each text query token position, select top-k heads by attention to image tokens; suppress all others with factor α. Per-position dynamic, no extra forward pass. |
| FlashVLM<br><br>FlashVLM: Text-Guided Visual Token Selection for Large Multimodal Models | 2512.20561 | arXiv Dec 2024 | LLaVA-1.5-7B<br><br>VQAv2, GQA, POPE, MME, MMBench | Query-conditioned token **pruning** (not boosting): cross-modal similarity between projected image tokens and text embeddings in LLM space → keep top-k most query-relevant tokens + non-redundant background set |
| CAAC<br><br>Mitigating Hallucination in LVLMs via Adaptive Attention Calibration | 2505.21472 | arXiv May 2025 | InstructBLIP, LLaVA-1.5, LLaVA-NeXT<br><br>CHAIR, POPE, AMBER | Two-step: (1) **VTC** smooths attention spikes using blank-image reference; (2) **AAR** scales image attention by `λ∝(1−confidence)` when model is uncertain. No layer/head selectivity, not query-conditioned. |
| Perception Magnifier | 2503.10183 | ACL 2026 | LLaVA-1.5<br><br>POPE, MME, | Iterative attention-based region magnification: aggregate self-attention |

| Paper | arXiv | Venue | Arch Tested | Core Idea |
|---|---|---|---|---|
| Through the Magnifying Glass: Adaptive Perception Magnification for Hallucination-Free VLM Decoding | | | LLaVA-Bench, MMHal | (L≥12) → mask dominant tokens → re-feed magnified overlooked regions. Query-conditioned (last-token attention). Shape distortion is a known side effect. |
| When to Think and When to Look<br><br>Uncertainty-Guided Lookback for Visual Grounding in Reasoning VLMs | 2511.15613 | CVPR 2026 | Qwen3-VL, InternVL3<br><br>MMMU, MMStar, MathVista | For CoT/reasoning VLMs: detect "lookback trigger phrases" during thinking chains; use Δ_content (PPL_Real−PPL_Noise) to score M re-grounding branches; select most visually-grounded path. −30% token use, +6pp MMMU. |
| Foveated Diffusion<br><br>Foveated Diffusion: Efficient Spatially Adaptive Image Generation | 2603.23491 | arXiv Mar 2026 | DiT-based diffusion<br><br>image/video generation | **Cross-domain inspiration:** Higher token density (foveal) in task-relevant regions, lower in periphery — spatially adaptive compute allocation. Directly motivates SRF's CLIP-guided non-uniform token boosting. |
| AdaVBoost<br><br>AdaVBoost: Mitigating Hallucinations in LVLMs via Token-Level Adaptive Visual Attention Boosting | 2602.13600 | arXiv Feb 2026 | LLaVA-NeXT-7B, Qwen3-VL-8B, InternVL3.5-8B<br><br>CHAIR, AMBER, SHR | Token-level adaptive attention boost during each generation step using Visual Grounding Entropy (VGE) — per-token risk estimate from prior step controls boost magnitude. Simultaneously suppresses text-token attention. |
| PADE<br><br>Revealing and Enhancing Core Visual Regions via Internal Attention Dynamics | 2602.15556 | arXiv Feb 2026 | LLaVA-1.5-7B, Qwen-VL<br><br>CHAIR, POPE, AMBER, MME | Constructs Positive Attention Dynamics (PAD) map from inter-layer attention *increases*; boosts final-layer attention logits for PAD-identified visual regions with per-head MAD scaling; compensates by reducing system-token attention (not instruction tokens). |

| Paper | arXiv | Venue | Arch Tested | Core Idea |
|---|---|---|---|---|
| Focus Matters<br><br>Focus Matters: Phase-Aware Suppression for Hallucination in VLMs | 2604.03556 | arXiv Apr 2026 | LLaVA-1.5-7B/13B<br><br>CHAIR, POPE, AMBER | **Vision encoder** (ViT) intervention, not LLM decoder: identifies "focus phase" layers (7–11) by attention concentration metric R; suppresses low-relevance patch tokens in those layers via DPP-selected mask. |
| CAST<br><br>CAST: Caption-Guided Visual Attention Steering for Object Hallucination | 2605.04641 | arXiv Jun 2026 | LLaVA-1.5-7B, 5 architectures<br><br>POPE, CHAIR, MMHal | Identifies top-100 "caption-sensitive" heads via SVM; computes shift vectors S from caption vs. non-caption output differences; injects α·S into head outputs at inference — steers toward captioning-style visual grounding. |
| ILVAD<br><br>Finding the Correct Visual Evidence via Inter-Layer Visual Attention Discrepancy | 2605.20965 | arXiv Jun 2026 | LLaVA-1.5-7B, 5 architectures<br><br>CHAIR, POPE, MMHal, MME | Builds saliency map from inter-layer attention discrepancy (layer-by-layer subtraction filters attention sinks); exponentially boosts evidence tokens (exp(α·attn)); adds text-grounding pass to tie strongly-connected text tokens to visual evidence. |

> **10 of 12 papers are training-free**. **Saliency-R1** (LoRA+GRPO) and **REVERSE** are the exceptions. FlashVLM/HiPrune/SparseVLM are efficiency-oriented (pruning) rather than hallucination-oriented.

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## AdaptVis

**arXiv:** 2503.01773 | **Date:** Mar 2025 | **Venue:** ICML 2025
**Authors:** Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, Manling Li

**Core Intuition**

> Think of attention as a spotlight. For spatial reasoning ("is the cup to the left of the plate?"), the spotlight must land on the *right* objects — not just shine brighter overall. The key insight: if the model is **confident** about its answer, its spotlight is probably already in the right place → sharpen it. If it's **uncertain**, the spotlight may be misplaced → broaden it to search wider.

**Problem:** VLMs fail at basic spatial reasoning ("left/right/above/below") not because they lack visual attention, but because their attention hits the *wrong spatial locations*. Images comprise ~90% of input tokens but get only ~10% of attention. Uniform attention boosts make things worse — they just amplify wherever the model already (wrongly) looks.

**Key Finding:** Through attention rollout analysis on 10,000+ samples, the authors show:

1. Correct answers correlate with attention *aligned to the actual object locations*
2. Middle layers (17–18 in LLaVA-1.5) are where this spatial alignment matters most
3. Generation probability (confidence) reliably predicts whether the attention is trustworthy

**Solution — AdaptVis (temperature scaling + confidence gate):**

```
If P(first_token) ≥ β  →  logits[image_cols] *= α   (α > 1, sharpen)
If P(first_token) <  β  →  logits[image_cols] *= α   (α < 1, broaden)
```

Applied to image-token attention logits at the answer generation step, uniformly across all heads.

**Details:**

- Training-free; operates at decoding time only
- $\alpha \in \{0.5, 0.8, 1.2, 1.5, 2.0\}$, $\beta \in [0.2, 0.65]$ — selected on 20% validation split per dataset
- Targets last input token's attention to image (the "query" that generates the answer)
- No head-level selectivity

**Arch:** LLaVA-1.5-7B, LLaVA-1.6-7B | **Datasets:** WhatsUp (Controlled_A/B, COCO/VG), VSR (1,223 pairs)

**Results:**

- LLaVA-1.5 Controlled_A: 60.3% → 84.9% (+24.6 pts)
- LLaVA-1.6 Controlled_A: 48.2% → **98.2%** (+50.0 pts!)
- VSR F1 LLaVA-1.5: +11.2 pts; LLaVA-1.6: +9.9 pts
- Real-world COCO/VG: modest +0.6–7.3 pts

**GAPS / Ideas for us:**

- Needs labeled validation set for α, β — not zero-shot portable
- All heads treated equally — VHR/VAF are more surgical
- Only tested on LLaVA; not Qwen2.5-VL (our model)
- β is dataset-specific; doesn't generalize across VLM Bias/POPE/MMBench without retuning
- **Direct connection:** This is the base of our `temperature` + `vision_boost` implementations. Our VLM Bias task is harder — language prior dominates, not just spatial misalignment

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## ClearSight

**arXiv:** 2503.13107 | **Date:** Mar 2025 | **Venue:** CVPR 2025
**Authors:** Hao Yin, Guangzong Si, Zilei Wang (USTC) | **Code:** github.com/ustc-hyin/ClearSight

> **Core Intuition**
>
> Imagine the transformer as a telephone relay — information passes through a chain of layers. In the *middle* of the chain (layers 8–15), visual and textual signals are being actively merged ("fusion"). At this stage, the model should be paying close attention to the image. But it turns out it's often paying *more* attention to the system prompt ("You are a helpful assistant...") than to the actual image content. ClearSight fixes this: in those critical middle layers, **turn up** the image attention and **turn down** the system-prompt attention — like adjusting equalizer sliders.

**Problem:** Methods like VCD fix hallucination by running a *second* inference pass with a distorted image and subtracting logits. This works but: (1) nearly doubles inference time, (2) degrades output quality because it suppresses ALL language knowledge including useful grammar/fluency (NoCaps CIDEr: 78.7 → 65.7).

**Key Finding:** Saliency analysis reveals middle transformer layers (8–15) have the highest visual-textual information flow (metric S_vt). In these layers, the model allocates *less attention to image tokens than to system-prompt tokens* — a measurable imbalance that correlates with object hallucination rate.

**Solution — Visual Amplification Fusion (VAF):**

The attention score matrix Z is modified only in layers 8–15, only for the top-50% vision-aware heads:

```
Ẑ_{l,h} = Z_{l,h}  +  α · M_enh ⊙ Z_{l,h}  −  β · M_sup ⊙ Z_{l,h}
```

Where:

- `M_enh` = binary mask selecting attention positions from **instruction tokens** → **image tokens** (rows=instruction queries, cols=image keys) → *enhance these*
- `M_sup` = binary mask selecting **instruction tokens** → **system-prompt tokens** → *suppress these*
- $\alpha=0.15$ (push up image attention by 15%), $\beta=0.1$ (pull down system attention by 10%)

Vision-aware heads selected as top 50% by mean image attention allocation (pre-computed once).

**Why only middle layers?** Early layers do low-level feature extraction (no semantic fusion yet). Late layers do output prediction (modifying here hurts fluency). Middle layers are the "decision zone" where visual context should be integrated into language generation.

**Details:**

- Overhead: <5% latency (no extra forward pass); same GPU memory
- Linear-projection architectures only (LLaVA-style); Q-Former (BLIP-2) not tested
- Optimal: $\alpha \in (0, 0.25)$, $\beta \in (0, 0.15)$

**Arch:** LLaVA-v1.5-7B/13B, Qwen-VL-7B | **Datasets:** POPE, MME, NoCaps, ScienceQA

**Results:**

| Method | POPE | MME (obj) | NoCaps CIDEr | ScienceQA |
|---|---|---|---|---|
| Baseline | ~87% | ~355 | 78.7 | 68.3% |
| VCD | +2% | +5% | **65.7** ↓ | 64.5% ↓ |
| **VAF** | **+3%** | **+7%** | **78.8** ✓ | **68.5%** ✓ |

**GAPS / Ideas for us:**

- Layer range 8–15 is hardcoded for LLaVA — Qwen2.5-VL has different fusion dynamics
- α, β are fixed; no adaptive mechanism per head/sample
- Not evaluated on spatial reasoning or bias tasks
- **Our extension (implemented):** `vaf` method in `qwen_attn_patch.py` — need to empirically find Qwen2.5-VL's fusion layers via saliency

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# VHR

**arXiv:** 2412.13949 | **Date:** Dec 2024 | **Venue:** ACL 2025
**Authors:** Jinghan He, Kuan Zhu, Haiyun Guo, Junfeng Fang, Zhenglin Hua, Yuheng Jia, Ming Tang, Tat-Seng Chua, Jinqiao Wang (NUS, Beijing Institute of Technology)

**Core Intuition**

A VLM's multi-head attention has many heads doing different jobs. Some heads have "learned to look at the image" (vision-aware), others mostly process language. Instead of boosting all heads equally (which amplifies language-biased heads too), find *which specific heads* are vision-sensitive and boost only those. It's like having 16 musicians — don't turn up all microphones, only the ones playing the visual melody.

To identify vision-aware heads: run the model twice — once with the image, once without. Heads whose *output changes a lot* between these two runs are the ones that actually use the visual input.

**Problem:** MLLMs produce object hallucinations because they over-rely on language priors ("a kitchen typically has a microwave" → hallucinates microwave). Boosting all attention heads uniformly (VCD, vision_boost) amplifies language-biased heads too, limiting effectiveness.

**Key Finding:** At each layer, attention heads vary enormously in how much they actually attend to image tokens. The top ~20% of heads have 3–5× higher visual attention than the bottom 20%. Scaling those top heads (VHR) gives better hallucination reduction than scaling all heads uniformly.

**Solution:**

1. **VHD (Vision-aware Head Divergence):** Score each head by how much its output changes when image is removed:
   ```
   VHD_h = || output_h(full_input) − output_h(text_only) ||_2
   ```
2. **VHR:** During generation step 0, scale the output of top-VHD heads by α=2:
   ```
   output_h_scaled = α · output_h   for high-VHD heads
   ```
   Applied to last 14 layers (LLaVA) or last 18 layers (InstructBLIP).

**Difference from our implementation:** VHR paper computes VHD per-sample (requires extra forward pass without image each time). Our implementation uses *calibration-based mean attention scoring* — runs once on N calibration samples, reuses the same head mask for all test samples. This is faster but less sample-specific.

**Details:**

- Training-free; extra cost = 1 forward pass per sample at step 0
- α=2 optimal across architectures; layers = last 50% of decoder
- Head selection frequency: top ~20% by VHD score

**Arch:** LLaVA-1.5-7B, InstructBLIP-7B | **Datasets:** CHAIR (MSCOCO 500), POPE, LLaVA-Bench

**Results:**

| Model | CHAIR_S ↓ | CHAIR_I ↓ | POPE F1 ↑ |
|---|---|---|---|
| LLaVA-1.5 baseline | 49.68 | 14.32 | ~85% |
| **+ VHR** | **33.32 (-32.9%)** | **9.71 (-32.2%)** | ↑ **consistently** |

**GAPS / Ideas for us:**

- VHD extra forward pass = doubles prefill cost per sample
- Fixed α=2 — could be adaptive per layer or head
- Only object hallucination; not tested on spatial or bias tasks
- Our calibration-based approach trades per-sample accuracy for speed — reasonable tradeoff
- **Implemented as `vhr_boost`** in our codebase; VHR head mask now uses text-query-only scoring (fixed in recent update)

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## VISTA

**arXiv:** 2502.03628 | **Date:** Feb 2025 | **Venue:** arXiv
**Authors:** Zhuowei Li, Haizhou Shi, Yunhe Gao, Di Liu, Zhenting Wang, Yuxiao Chen, Ting Liu, Long Zhao, Hao Wang, Dimitris N. Metaxas (Rutgers University)

> **Core Intuition**
>
> Think of each layer's hidden state as carrying a "memory" of what the model has seen. As generation proceeds, visual information *fades* — the model's internal representation drifts toward linguistic patterns. VISTA works like two corrective interventions:
>
> 1. **VSV (steering vector):** At each step, compute the direction in hidden-state space that corresponds to "having visual context" (image − no-image), then push the hidden state a little in that direction. Like correcting a drifting boat back on course.
> 2. **SLA (logit blending):** The final layer's logits are often over-refined toward language patterns. Earlier layers still have strong visual signal. Average the logits from the last

few layers and blend them with the final output — bring back the visual signal that was refined away.

**Problem:** Visual hallucination happens progressively during generation. The model starts grounded but drifts toward language priors. Token logit analysis shows: genuine visual tokens *lose ranking* across layers, while hallucinated tokens surface from language statistics.

**Key Finding (from logit rank tracking):**

- Correct tokens peak in probability at *penultimate* layers (not the final layer)
- The final layer has actually *over-processed* and drifted away from visual grounding
- The gap between "what should be generated" and "what is generated" grows with generation length

**Solution:**
**VSV — Visual Steering Vector:**

```
v_steer^l = h^l(with_image) − h^l(without_image)   # computed once per sample
h̃_t^l = h_t^l + λ · normalize(v_steer^l)          # injected at each step
```

**SLA — Self-Logits Augmentation:**

```
o_aug = mean(o^{L-k}, ..., o^{L-1})   # avg logits from preceding k layers
õ_t = (1−γ) · o_t^L + γ · o_aug       # blend with final-layer logits
```

Both components add together; VSV corrects the hidden state, SLA corrects the output distribution.

**Details:**

- 1.27× latency overhead (extra forward passes for VSV, early-layer logit extraction for SLA)
- $\lambda$ (VSV strength) and $\gamma$ (SLA blend) vary per architecture — needs validation set
- Works on multiple VLM families (LLaVA, MiniGPT-4, InstructBLIP)

**Arch:** LLaVA-1.5, MiniGPT-4, InstructBLIP | **Datasets:** MSCOCO 500 (CHAIR), POPE, MMHal-Bench, MME

**Results:**

| Model | CHAIR_S ↓ | POPE ↑ | MME ↑ |
|---|---|---|---|
| LLaVA-1.5 baseline | 46.4 | 86.1% | — |

| **+ VISTA** | **20.4 (-56%)** | **86.2%** | — |
| MiniGPT-4 baseline | — | — | ~600 |
| **+ VISTA** | — | — | **+224 pts** |

**GAPS / Ideas for us:**

- VSV requires extra forward pass without image (same cost as VHR's VHD)
- λ, γ are architecture-specific; portability requires per-model validation
- Embedding space misalignment: applying final layer's projection to earlier hidden states introduces error
- **Idea:** SLA alone (logit blending) is cheap — no extra forward pass, just hook hidden states with `output_hidden_states=True`. Good candidate for quick implementation

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# VAR

**arXiv:** 2503.03321 | **Date:** Mar 2025 | **Venue:** arXiv
**Authors:** Seil Kang, Jinyeong Kim, Junhyeok Kim, Seong Jae Hwang (Yonsei University)

**Core Intuition**

In language models, certain early tokens (like the first token or separator tokens) act as "attention sinks" — they collect attention from all other tokens, not because they're informative but because the model uses them as a neutral "garbage collector." The same phenomenon happens with *visual* tokens in VLMs: certain fixed spatial positions in the image always receive high attention regardless of the image content or the question. This wastes the model's attention budget. VAR detects these visual attention sinks and **redirects** their stolen attention to visual tokens that actually contain relevant content.

**Problem:** In VLMs, specific visual token positions (often corners or fixed patches) consistently attract disproportionate attention — not because they're semantically relevant, but due to massive activations in specific hidden dimensions inherited from the LM backbone. This wastes attention capacity that should go to semantically important image regions.

**Key Finding:** Visual attention sinks are caused by *massive activation* in specific hidden-state dimensions (same as LM attention sinks in StreamingLLM / SinkCache). These dimensions can be detected by checking which dimensions have very large normalized activations ϕ(x). Crucially, heads vary: "image-centric heads" have a high fraction of attention on *non-sink* visual tokens — these are the heads worth fixing.

**Solution — Visual Attention Redistribution (VAR):**

1. **Identify sink dimensions:** `ϕ(x) = max_d(|x_d| / ||x||)` — dimensions where activations dominate
2. **Identify image-centric heads:** heads where visual non-sink ratio `r^{l,h} ≥ ρ` (i.e., heads that attend to real visual content, not just sinks)
3. **Redistribute:** For image-centric heads, take fraction p=0.6 of the attention assigned to sink tokens and redistribute it proportionally to non-sink visual tokens

```
# For image-centric heads only:
attn_sink_portion = p * attn[:, sink_tokens]     # steal 60% from sinks
redistribute to non-sink visual tokens ∝ their current attention
```

**Details:**

- Post-hoc; no retraining; sink dimensions identified once from base LM (offline)
- Threshold ρ varies by task type (0.5–0.9); catastrophic if applied to all heads
- Tested on 6 VLM families including Qwen2-VL ✓

**Arch:** LLaVA-1.5/1.6, VILA, Qwen2-VL, InternVL2 | **Datasets:** VQAv2, GQA, VizWiz, MME, MMBench, SEED, MMVP, CHAIR, POPE, MMHal-Bench, CV-Bench

**Results:**

- MMVP (vision-centric): 3.33 → **9.33** (+7 pts on LLaVA-1.5-7B) — biggest gain
- CHAIR_S: 45.0 → 43.2 (modest)
- POPE F1: 85.9 → 86.5
- General VQA tasks: +1–3% across models

**GAPS / Ideas for us:**

- ρ threshold requires task-specific tuning — not plug-and-play
- Sink dimension identification from base LM may not transfer to all architectures
- Biggest gains on spatial/visual tasks (MMVP); smaller on MCQ/yes-no (our tasks)
- **Idea:** Combine with VHR — VHR selects vision-aware heads, VAR fixes attention sinks within those heads; could compound gains

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## Localization Heads

**arXiv:** 2503.06287 | **Date:** Mar 2025 | **Venue:** arXiv
**Authors:** Seil Kang, Jinyeong Kim, Junhyeok Kim, Seong Jae Hwang (Yonsei University)

### Core Intuition

When you ask "where is the dog?" and the model answers correctly, *some* attention heads must be doing the spatial work — computing which image patch corresponds to "dog." The authors ask: can we find *which* heads those are? It turns out only **~3 heads out of hundreds** consistently act as spatial localization specialists across all queries. Remarkably, you can extract bounding boxes from these 3 heads alone — without any fine-tuning on localization data — by simply reading off their attention maps.

**Problem:** Visual grounding (producing bounding boxes or segmentation masks from text descriptions) requires expensive fine-tuning with specialized datasets in methods like LISA, Ferret, Shikra. Can frozen VLMs already do this with the right attention heads?

**Key Finding:** Not all attention heads attend to image tokens in a spatially meaningful way. Most heads distribute attention broadly or focus on language context. A tiny subset of heads:

- Attend heavily to image tokens overall (high **attention sum**)
- Concentrate their attention in a localized spatial region (low **spatial entropy**)

These "localization heads" are consistent across different queries and images — they are structural properties of the model, not query-specific.

**Solution:**

1. **Score each head** across 1,000 calibration samples: attention sum + spatial entropy
2. **Select top-3 heads** by how often they rank in the top-k (selection frequency)
3. **At inference:** Take the last text token's attention map for these 3 heads → average → Gaussian smooth (k=7, σ=1.0) → threshold → convex hull → bounding box

```
# Localization pipeline (inference only, no gradient):
attn_maps = [attention[head_i][last_text_token → image_tokens] for head_i in top_3]
heatmap = sum(attn_maps)
heatmap = gaussian_smooth(heatmap, k=7, σ=1.0)
bbox = convex_hull(heatmap > threshold)
```

**Details:**

- Calibration: 1,000 samples from training split of RefCOCO (but no gradient descent)
- Threshold τ via maximum curvature heuristic (automatic)
- Performance scales with model size (1.3B → 7B → 13B)

- Only 3 heads needed — remarkably sparse

**Arch:** LLaVA-1.5 (1.3B, 7B, 13B) | **Datasets:** RefCOCO, RefCOCO+, RefCOCOg, ReasonSeg

**Results:**

| Model | REC (val) | RES cIoU | Comparable to |
|---|---|---|---|
| LLaVA-1.5-13B | **87.2%** | **76.1%** | LISA-7B (fine-tuned) |
| LLaVA-1.5-7B | 82.4% | 70.3% | Ferret-7B (fine-tuned) |

**GAPS / Ideas for us:**

- Head selection uses calibration data — not purely zero-shot
- Fails when multiple instances of same category present (no disambiguation)
- Not tested on Qwen2.5-VL or dynamic-resolution / tiling models
- **Diagnostic use for us:** Apply to VLM Bias samples — visualize which image regions the model is attending to when it gives a biased answer. Does it look at the biasing object or the question-relevant object?

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# OPERA

**arXiv:** 2311.17911 | **Date:** Nov 2023 | **Venue:** CVPR 2024
**Authors:** Qidong Huang, Xiaoyi Dong, Pan Zhang, Bin Wang, Conghui He, Jiaqi Wang, Dahua Lin, Weiming Zhang, Nenghai Yu (USTC, Shanghai AI Lab, CUHK)

> **Core Intuition**
>
> When a VLM writes a long description, it sometimes enters a "lazy mode" — instead of going back to the image for each new clause, it lets a single summary token (a period, a comma, "and") accumulate all the context, then predicts the next word from that summary token alone. This produces fluent but hallucinated text. OPERA detects this "columnar" attention pattern (many tokens all attending to one summary token) and either penalizes it or makes the model backtrack and try a different path.

**Problem:** MLLMs produce long descriptions with hallucinated objects/attributes. The root cause: in self-attention maps, certain tokens (punctuation, line breaks, "\n\n") act as "summary aggregators" — they concentrate all previous context. When the model over-relies on these summaries, it loses track of what's actually in the image.

**Key Finding:** Hallucinated content in long generations correlates with "columnar" attention patterns — many tokens attending to a single summary position, creating a column of high attention values in the attention map. The model stops attending to visual tokens and instead recycles its own compressed summary.

**Solution:**

1. **Over-Trust Penalty:** At each beam search step, detect if a columnar pattern is forming:
   ```
   p(x_t|x<t) = Softmax[ H(h_t)  −  α · ϕ(w_≤t) ]
   ```
   Where `ϕ(w_≤t)` measures the column-wise attention aggregation strength.
2. **Retrospection-Allocation:** If the pattern persists for r=15 consecutive tokens, roll back decoding and select from N_can=5 alternative candidates.

**Details:**

- Built on beam search (beam=5) — not applicable to greedy/sampling
- Hyperparameters: σ=50, α=1, r=15, N_can=5 (architecture-specific tuning needed)
- Works best on long-form descriptions (>50 tokens); minimal gains on short answers

**Arch:** InstructBLIP-7B, MiniGPT-4-7B, LLaVA-1.5-7B, Shikra-7B | **Datasets:** MSCOCO (CHAIR), Visual Genome, MMBench, MME

**Results:**

- Shikra CHAIR_S: 50.4 → 36.2 (-28%); GPT-4 eval: +27.5% correctness
- POPE: marginal (+0.4 F1) — expected, since POPE is yes/no (short answer)

**GAPS / Ideas for us:**

- Beam search only — our eval uses greedy decoding → not applicable
- Short-answer tasks (VLM Bias, POPE, MMBench) get negligible benefit
- Cannot fix hallucinations caused by strong training priors
- **Not useful for our current tasks; useful for captioning/VQA evaluation**

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## LLMind

**arXiv:** 2603.14882 | **Date:** Mar 2026 | **Venue:** arXiv
**Authors:** Soumyaratna Debnath, Bui Duc Manh, Zinan Liu, Lin Wang (NTU Singapore)

**Core Intuition**

Human eyes don't process all parts of an image equally — the fovea (center of gaze) processes in high detail, the periphery in low detail. When you look at a photo to answer "what is the man holding?", your eyes zoom in on the man's hands, compressing the background. LLMind mimics this: mathematically warp the image so that the "important" region is magnified and the background is compressed, then feed this warped image to the VLM. The "importance" is determined by optimizing for the correct answer — using a gradient-free method (SPSA) since many VLMs are black boxes.

**Problem:** VLMs process images as a fixed grid of patches — all patches get equal computational budget. Irrelevant background patches consume the same attention and token budget as the foreground object, creating noise and reducing effective visual resolution on what matters.

**Key Finding:** At 1–5% of original pixel budget (simulating limited resolution), biologically-inspired non-uniform sampling outperforms uniform sampling by +20–38% on VQA. The key is *where* you spend your pixel budget, not how many pixels you have.

**Solution — BASS (Bio-inspired Adaptive Sampling Strategy):**

1. **Möbius Transformation:** A conformal map (angle-preserving warp) parameterized by $\theta \in \mathbb{R}^4$ that stretches the image around a fixation point while compressing the periphery. Applied via stereographic projection to preserve smooth topology.
2. **Closed-Loop Semantic Feedback (CSF):** Optimize θ to maximize task performance:

   ```
   ℒ_total = α·ℒ_VSI + β·ℒ_DISTS + γ·ℒ_MSE     # perceptual loss (image quality)
            + δ·(1 − cosine(embed(y_pred), embed(y_gt)))  # semantic loss
   ```

   Optimized via **SPSA** (Simultaneous Perturbation Stochastic Approximation) — estimates gradient by perturbing θ by ±Δ and comparing task losses. Works on black-box VLMs.
3. **Adaptive example selection:** Focus optimization effort on hard examples using exponential weighting by error rate.

**Details:**

- Training-free; SPSA adds ~4.75s/iteration overhead
- Requires ground-truth labels for optimization (test-time adaptation setting)
- Tested on 5 VLM families including Qwen2.5-VL-4B ✓

**Arch:** Qwen2.5-VL-4B, SmolVLM-2B, LLaVA-OneVision-0.9B, Qwen3-VL-2B, DeepSeek-VL-2B

**Datasets:** VQAv2, Seed-Bench, A-OKVQA, LVIS (region-guided VQA)

**Results:**
| Dataset | 1% pixels | 3% pixels | 5% pixels | Full res |
|---|---|---|---|---|
| VQAv2 | +20% vs uniform | +15% vs uniform | +10% vs uniform | — |
| Seed-Bench | +38% | +25% | +18% | — |
| Retains | 82% full-res | 92% full-res | 97% full-res | 100% |

**GAPS / Ideas for us:**

- SPSA overhead (4.75s/iter) makes it slow for large-scale eval
- Needs ground-truth for optimization — semi-supervised setting
- Single static warp; no sequential fixation like real saccades
- **Hypothesis for VLM Bias:** Model attends to biasing background objects → warping to suppress background could reduce bias. Direct experiment: apply BASS with fixation on the question-relevant object

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## AIR

**arXiv:** 2602.24041 | **Date:** Feb 2026 | **Venue:** ICLR 2026
**Authors:** Xingyu Zhu, Kesen Zhao, Liang Yi, Shuo Wang, Zhicai Wang, Beier Zhu, Hanwang Zhang, Xiangnan He (USTC, NTU Singapore)

**Core Intuition**

Instead of boosting all visual tokens uniformly (which amplifies noisy background patches as much as foreground), AIR first compresses the token pool to a distinctive subset via a global prototype, then uses Optimal Transport to find which patches align with the model's own hidden representation at the FFN step. Only those best-aligned patches get re-injected — like handing the model a curated highlight reel instead of the whole film. The key difference from attention-level methods: AIR operates *after* attention, in the FFN computation, on the value content rather than the routing weights.

**Problem:** Indiscriminate visual token boosting (VCD, VHR, ClearSight) amplifies background regions as much as foreground, adding noise and diluting signal from semantically relevant image patches.

**Key Findings:**

1. Salient image patches show consistently higher alignment with the model's hidden states than background patches
2. OT-based selection creates larger discriminative margins between salient and non-salient tokens than cosine averaging (proven theoretically)
3. Optimal intervention layers are the **last third of the decoder** (layers 24–32 for LLaVA-1.5) — unlike ClearSight's middle layers

**Solution — AIR (Adaptive vIsual Reinforcement):**

Operates inside transformer **FFN layers** (not attention):

```
Step 1 — Prototype token reduction:
    prototype = mean(all visual token embeddings)    # coarse visual summary
    scores = L2_distance(each_token, prototype)
    keep top-Q most distinctive tokens               # Q=100

Step 2 — OT-guided patch reinforcement:
    For each kept token, compute OT distance to image patch embeddings
    via Sinkhorn-Knopp algorithm
    Select tokens where OT_distance ≤ τ             # τ=0.06
    Re-inject selected patch embeddings into FFN output residual
```

**Details:**

- Training-free; operates during forward pass inside FFN, not attention
- TopQ=100, τ=0.06, patch_count=12; optimal layers: last third of decoder
- Latency: +0.4s per sample (2.07s vs 1.68s baseline on single A100)
- Greedy decoding (temp=0); tested on 3 architectures

**Arch:** LLaVA-1.5-7B, Qwen-VL-Chat, GLM-4V-9B **| Datasets:** CHAIR (MSCOCO 500), POPE (random/popular/adversarial), HallusionBench, V*Bench, MMHal-Bench, MM-Vet, MME, MMBench

**Results (LLaVA-1.5-7B):**

| Method | CHAIR_S ↓ | CHAIR_I ↓ | POPE random ↑ | POPE adversarial ↑ |
|---|---|---|---|---|
| Baseline | 22.0 | 6.7 | 83.7% | 75.0% |
| VAF | 20.4 | 6.5 | 87.6% | 83.9% |
| **AIR** | **18.4** | **5.7** | **89.0%** | **83.9%** |

**GAPS / Ideas for us:**

- AIR's token selectivity is **query-agnostic**: it selects distinctive tokens relative to the visual prototype regardless of what the question asks. Two questions on the same image → same tokens selected. **SRF's CLIP/HSSA saliency is query-conditioned** — "Is there a chair?" and "Is there a clock?" select different tokens on the same image. This is the key differentiator in our novelty table.
- AIR acts in FFN layers; SRF acts in attention layers — complementary intervention sites (could be combined for cumulative gain)
- Not tested on Qwen2.5-VL (our primary model)
- On POPE adversarial, AIR matches VAF (both at 83.9%) — our SRF adds query-conditioned token selectivity on top of layer + head targeting, targeting a further improvement here
- **Novelty comparison:** AIR = token selectivity in FFN (unsupervised). SRF = token selectivity in attention (query-supervised). Different domain, different conditioning signal — distinct contribution.

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## Saliency-R1

**arXiv:** 2604.04500 | **Date:** Apr 2026 | **Venue:** CVPR 2026
**Authors:** Shizhan Gong, Minda Hu, Qiyuan Zhang, Chen Ma, Qi Dou (CUHK, CityU Hong Kong)
**Code:** github.com/peterant330/Saliency_R1

> **Core Intuition**
>
> ⚠️ Unlike every other paper in this list, Saliency-R1 is **not training-free** — it uses GRPO reinforcement learning to teach the model to attend to the right image regions. The saliency component is a *measurement* tool: decompose exactly which image patches contributed to each output token via logit arithmetic, then reward the model for attending to regions that overlap with ground-truth bounding boxes. The result is a model that has *learned* better visual grounding — not an inference-time patch. Think of it as SRF's trained counterpart: we correct at inference time, they bake the correction into weights.

**Problem:** VLMs over-rely on textual cues and hallucinate. Standard saliency methods (attention rollout, Grad-CAM) don't faithfully reflect what the model *actually uses* — they're approximations. Without faithful saliency, you can't reward the model for correct visual grounding during training.

**Key Findings:**

1. Token logits can be exactly decomposed into first-order contributions from each context token (image patches included) — purely via matrix operations, no extra forward pass
2. Tracing this saliency through "thinking tokens" (chain-of-thought steps) reveals whether reasoning genuinely uses visual content or just rephrases language priors
3. GRPO training with this saliency-overlap reward produces models with the lowest deletion scores (i.e. the highlighted regions are the most causally important to the output)

**Solution:**

**Component 1 — Token-level saliency via logit decomposition:**
Decomposes each output token's logit into exact first-order contributions from every context token (image patches). Implemented as pure matrix operations — no extra forward/backward pass.

**Component 2 — Holistic aggregation with thinking bottleneck:**
Uses attention rollout to trace information flow through CoT reasoning tokens: image → thinking tokens → answer. Reveals whether the reasoning chain is genuinely grounded in the image.

**Component 3 — GRPO training:**

```
reward = IoU(saliency_map, GT_bounding_boxes)
```

SFT warm-up on 272K samples (Vision-R1-cold), then GRPO with LoRA (rank=16) on 8,080 annotated VQA samples from Visual-CoT (OCR, fine-grained, charts, relation reasoning).

**Details:**

- **Requires LoRA fine-tuning + GRPO training** — not training-free
- Training data needs GT bounding-box annotations (8K samples)
- Saliency computation at inference: pure matrix ops, no overhead
- GPT-4o-mini used as judge for accuracy; llama-factory + TRL for training

**Arch:** Qwen2.5-VL-3B-Instruct, Qwen2.5-VL-7B-Instruct | **Datasets:** COCO Caption, OpenPSG, GranDf (saliency eval); Visual-CoT (training); downstream VQA tasks

**Results — saliency faithfulness (Qwen2.5-VL-3B, COCO Caption):**

| Method | Deletion@30% ↓ | Insertion@30% ↑ | Requires training? |
|---|---|---|---|
| Attention rollout | 58.48 | 38.43 | No |
| ATTN-LRP | 52.92 | 45.67 | No |
| TAM | 73.29 | 45.24 | No |
| **Saliency-R1** | **50.34** | **45.22** | **Yes (LoRA+GRPO)** |

Lower deletion = removed highlighted pixels hurt performance most (causally important); higher insertion = highlighted pixels alone recover most performance.

**GAPS / Ideas for us:**

- **Fundamentally different paradigm from SRF**: Saliency-R1 requires fine-tuning; SRF is training-free. These don't compete — Saliency-R1 is the trained upper bound on what's achievable when you can afford supervision.
- Both address the same root failure (model ignores query-relevant visual regions) but from opposite ends: SRF corrects attention routing at inference; Saliency-R1 changes what the model has learned to attend to.
- **Their logit decomposition saliency is a useful diagnostic for us**: we can apply it (training-free component only) to visualize which image tokens actually contributed to the model's answer before and after SRF intervention — a concrete interpretability figure for the paper.
- Tested on Qwen2.5-VL (our model) — their GRPO-trained numbers give a supervision upper bound for our benchmarks.
- **Natural follow-up**: SRF identifies the right tokens at inference; Saliency-R1 shows a path to teaching the model to find them during training. Our SRF saliency mask (CLIP/HSSA) could serve as the reward signal — removing the need for GT bounding boxes.
- Their SFT+GRPO pipeline requires annotated bounding boxes; SRF is annotation-free (CLIP similarity or internal hidden-state cosine alignment).

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## SPIN

**arXiv:** 2505.16411 | **Date:** May 2025 | **Venue:** arXiv
**Authors:** Sreetama Sarkar, Yue Che, Alex Gavin, Peter A. Beerel, Souvik Kundu

> **Core Intuition**
>
> VHR identifies vision-aware heads and *boosts* them. SPIN takes the opposite view: identify heads that are *ignoring* the image (low attention to visual tokens for the current query position) and *suppress* them. The analogy: instead of amplifying the musicians playing the visual melody, mute the ones drowning it out with language noise.

**Problem:** Hallucination from heads that consistently ignore image tokens during text generation, amplifying language priors over visual grounding.

**Solution:** For each text query token position, SPIN:

1. Computes each head's cumulative attention to vision tokens at that position
2. Selects top-k heads (those that *do* attend to image) → give them weight 1
3. Suppresses all other heads by factor $\alpha \in [0, 1]$

**Key distinction from VHR/ClearSight:** Per-position dynamic (head selection changes at each token position), not a fixed calibration-based mask. Suppression rather than boosting. No extra forward pass.

**Datasets:** POPE, CHAIR (COCO 2014 500 imgs), MMHal-Bench, MME, MMMU
**Results:** Up to 2.7× CHAIR hallucination reduction, 1.8× throughput improvement, ~7% GPT-4o accuracy gain (LLaVA-1.5-7B)

**GAPS / Ideas for us:**

- Not query-conditioned at the image-token level (doesn't select *which* image tokens to boost)
- Not tested on MMVP, VLM Bias, or Qwen2.5-VL
- Could combine: SPIN-style head suppression + SRF-style CLIP saliency mask = both dimensions
- **Cite in comparison table** — closest cousin to our head-selection approach; SRF adds the token-selectivity axis that SPIN lacks

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## FlashVLM

**arXiv:** 2512.20561 | **Date:** Dec 2024 | **Venue:** arXiv
**Authors:** Kaitong Cai, Jusheng Zhang, Jing Yang, Yijia Fan, Pengtao Xie, Jian Wang, Keze Wang

> **Core Intuition**
>
> If you know which image tokens are relevant to the question (via cross-modal similarity in LLM space), why feed the rest to every transformer layer? Keep only the relevant + a non-redundant background set, discard the rest. Different questions → different kept tokens. This is efficiency-focused but validates that query-conditioned token selection is meaningful.

**Problem:** VLMs process all image tokens at equal computational cost even when most are irrelevant to the question. Token redundancy wastes attention budget and can add noise.

**Solution — Two-stage token selection:**

1. **Relevance scoring:** Compute cross-modal cosine similarity between projected image token embeddings and normalized text embeddings *in LLM space* (after the vision encoder). Fuse with intrinsic visual saliency (log-domain weighting).
2. **Diversity preservation:** Bipartite residual pruning — keeps both high-scoring tokens AND a non-redundant background set (to avoid missing all-salient-region images).

**Query-conditioned:** Yes — Figure 3 explicitly shows different questions on the same image → different kept tokens.

**Datasets:** VQAv2, GQA, VizWiz, TextVQA, POPE, MME, MMBench, MMBench-CN, MMVet, TGIF-QA, MSVD-QA
**Results:** 77.8% pruning (128→16 tokens) achieves 100.6% relative accuracy on VQAv2 (exceeds unpruned baseline); 94.4% pruning (32 tokens) → 92.8% accuracy.

**GAPS / Ideas for us:**

- Efficiency goal (token pruning), not hallucination grounding goal — different problem but same insight
- Single-shot selection, no iterative refinement
- Cross-modal similarity is in LLM projection space (not external CLIP) — could be noisier for absent objects
- **Key insight for SRF:** FlashVLM validates query-conditioned token selection in LLM space. Our CLIP approach is external but more spatially interpretable. Consider adding LLM-space similarity as an alternative saliency signal (no CLIP overhead).

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# CAAC

**arXiv:** 2505.21472 | **Date:** May 2025 | **Venue:** arXiv
**Authors:** Mehrdad Fazli, Bowen Wei, Ahmet Sari, Ziwei Zhu | **Code:** github.com/mehrdadfazli/CAAC

> **Core Intuition**
>
> Two sources of attention imbalance: (1) fixed spatial attention spikes (some image patches always get high attention regardless of content), (2) confidence-driven language drift (when the model is uncertain, it falls back on language priors more). CAAC treats these separately with two cheap interventions.

**Problem:** Spatial perception bias (attention spikes to irrelevant patches) AND modality bias (language prior dominates when model is uncertain).

**Solution:**

- **VTC (Visual-Token Calibration):** Compute attention distribution on a blank image reference → get the "structural bias" pattern. Smooth actual attention toward blank-image pattern: `Vs = (1−β)V + βVu`. Subtracts the positional/structural artifact.
- **AAR (Adaptive Attention Re-Scaling):** Monitor generation confidence (max logit). When `p < 0.25` (uncertain), apply scaling `λt = λmin·p + λmax·(1−p)` with λmin=1, λmax=1.5 to image attention. More boosting when more uncertain.

**Not query-conditioned:** Both steps are image-only (VTC) or confidence-only (AAR) — the question doesn't change which tokens get boosted.

**Datasets:** POPE (adversarial/popular/random), CHAIR, AMBER | **Arch:** InstructBLIP, LLaVA-1.5, LLaVA-NeXT
**Results:** LLaVA-1.5: CHAIR_I 17.6→10.4 (−41%), POPE avg ~85.1%. Modest POPE gains.

**GAPS / Ideas for us:**

- VTC's blank-image reference is similar to SRF-E's no-image pass but used differently (calibration vs. contrastive)
- AAR's confidence gate is similar to AdaptVis — but applied to modality balance, not spatial sharpening
- **Idea for SRF:** Combine AAR's confidence gate with SRF's saliency boost — when confidence is low, apply LARGER alpha (model is unsure → push harder toward the visual evidence). Currently SRF's alpha is fixed.
- Not tested on MMVP or VLM Bias

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## Perception Magnifier

**arXiv:** 2503.10183 | **Date:** Mar 2025 | **Venue:** ACL 2026
**Authors:** Shunqi Mao, Chaoyi Zhang, Weidong Cai (University of Sydney)

> **Core Intuition**
>
> When you read a document and miss something, you go back and look more carefully at the part you skimmed. The Perception Magnifier does this with image regions: after a forward pass, extract the model's attention heatmap, mask the dominant (already-

attended) regions, find the *overlooked* patches, then magnify those patches and re-feed them. Iteratively surfaces what the model was ignoring.

**Problem:** Hallucination from model over-attending to easy/salient image regions and ignoring fine-grained or background-embedded visual details.

**Solution — Three-stage iterative pipeline:**

1. **Perception map:** Aggregate self-attention across L≥12 layers (max pooling over heads), apply variance amplification (α=10), smooth (k=3), upsample to pixel space → get current attention heatmap
2. **Iterative masking:** 2-means cluster on attention tokens; mask dominant cluster; repeat until total attention < β=0.3 threshold → reveals *neglected* regions
3. **Spatial magnification:** Inverse transform sampling → magnify the overlooked region; re-feed to model

**Query-conditioned:** Yes (last-generated-token attention drives the heatmap; changes per decoding step).

**Datasets:** POPE (86.70% vs 84.59% baseline), MME Perception (683→630), LLaVA-Bench, MMHal-Bench
**Key limitation:** Shape distortion from spatial magnification limits geometric reasoning. Breaks KV-cache efficiency (extra forward passes).

**GAPS / Ideas for us:**

- Input modification (image warp) vs. SRF's in-attention logit boost — different intervention point
- Shape distortion is the main practical limitation; SRF avoids this entirely (no image modification)
- Their iterative attention masking logic (find what the model is *already ignoring* → boost that) is complementary to CLIP saliency: **idea — use Perception Magnifier-style attention masking to find neglected query-relevant patches, then apply SRF boost to those** (two-signal saliency)
- Not tested on MMVP, VLM Bias, or Qwen2.5-VL

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## When to Think and When to Look

**arXiv:** 2511.15613 | **Date:** Nov 2025 | **Venue:** CVPR 2026
**Authors:** Jing Bi, Filippos Bellos, Junjia Guo, Yayuan Li, Chao Huang, Luchuan Song, Susan

Liang, Zhongfei Zhang, Jason Corso, Chenliang Xu

### Core Intuition

For reasoning VLMs (those that generate explicit thinking chains like Qwen3-VL), more thinking is not always better — long chains drift away from the image. This paper detects *when* the model has lost visual grounding mid-chain (via Δ_content = PPL_Real − PPL_Noise) and inserts explicit "lookback" phrases to force the model back to the image.

**Problem:** Chain-of-thought reasoning in VLMs improves language tasks but *hurts* visual reasoning — the model talks itself into a wrong answer by following textual logic and ignoring the image.

**Solution — Three components:**

1. **Visual grounding probe:** Two metrics per token position:
   - `Δ_content = PPL_Real − PPL_Noise` : how much correct image content aids prediction (negative = well-grounded)
   - `Δ_presence = PPL_Noise − PPL_Empty` : effect of merely having visual tokens present (decouples presence from content)
2. **Trigger mining:** Mine n-grams (pause phrases like "hmm", "wait") that correlate with low grounding. Insert "Looking back at the image..." when triggered.
3. **Breadth search:** Sample M continuations at trigger points; score by `V = −mean(Δ_content)` ; keep most visually-grounded branch.

**Different paradigm from SRF:** Targets CoT/thinking VLMs. Our work targets attention-level grounding in single-pass models. These are complementary.

**Datasets:** MMMU, MMStar, MathVista, MathVision, MathVerse | **Arch:** Qwen3-VL (4B/8B/32B), InternVL3
**Results:** +6.4 pts on MMMU-32B (75.3→81.7%), −30% token usage.

**GAPS / Ideas for us:**

- **Δ_content as SRF diagnostic:** Their metric (PPL_Real − PPL_Noise) could be computed *before* and *after* SRF intervention to quantify how much SRF improves visual grounding. A concrete interpretability contribution.
- **Adaptive α idea:** Use Δ_content per sample as a signal for how strongly to apply SRF. Low visual grounding (high Δ_content) → apply stronger boost. High grounding → lighter boost.
- Not applicable to our current eval (POPE, MMVP, VLM Bias are short-answer, non-CoT tasks)

## Foveated Diffusion

**arXiv:** 2603.23491 | **Date:** Mar 2026 | **Venue:** arXiv
**Authors:** Brian Chao, Lior Yariv, Howard Xiao, Gordon Wetzstein

> **Cross-Domain Inspiration**
>
> Diffusion models allocate tokens uniformly across the image. Human vision allocates resolution non-uniformly (fovea = high res center, periphery = low res). Foveated Diffusion builds mixed-resolution token grids — more tokens where detail matters, fewer in the periphery — achieving perceptually indistinguishable output with far fewer tokens. The key concept: **spatially non-uniform compute allocation driven by semantic relevance**.

**Core concept transferable to SRF:**

- **Foveated Diffusion:** Dense tokens at fixation point (known/estimated gaze) → efficient high-res generation
- **SRF analogue:** Dense *attention weight* at CLIP-salient image tokens → efficient visual grounding

The Foveated Diffusion → SRF mapping:

- "Fixation point" = CLIP saliency map (query-conditioned noun relevance)
- "Token density gradient" = attention logit boost gradient (soft mask, not binary)
- "Mixed-resolution grid" = non-uniform salience weights applied to attention logits

**Diffusion CFG → SRF-E analogy:**
Classifier-free guidance runs two passes (conditional + unconditional) and amplifies the difference. SRF-E does exactly this: `logits_final = logits_full + β(logits_full − logits_noval)`. The improvement frontier: instead of zeroing all pixel_values (fully unconditional), use a *spatially masked* version (zero only the salient region) → contrast specifically measures what the relevant region contributes.

**GAPS / Ideas for us:**

- **Spatial CFG for SRF-E:** `null_input = image with CLIP-salient patches zeroed` → `logits_spatial_cfg = logits_full + β(logits_full − logits_masked_region)`. Isolates the salient region's contribution. Fixes VLM Bias format-token suppression (null input still has format context).

- **Soft foveal gradient:** Instead of binary threshold (mask or no mask), use a smooth saliency gradient for logit boost (already partially done with soft mask in SRF). Foveated Diffusion's smooth resolution falloff is the analogue.

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## AdaVBoost

**arXiv:** 2602.13600 | **Date:** Feb 2026 | **Venue:** arXiv
**Authors:** Jiacheng Zhang, Feng Liu, Chao Du, Tianyu Pang

> **Core Intuition**
>
> Every generation step is a different risk level. If the model is already well-grounded (low hallucination risk at step t), a strong attention boost wastes capacity and can distort fluent tokens. If the model is about to hallucinate (high risk), you want maximum boost. AdaVBoost measures hallucination risk *per token* using Visual Grounding Entropy (VGE — predictive uncertainty weighted by visual grounding signal), then scales the boost proportionally. Like a thermostat: turn up the heat only when it's cold.

**Problem:** Fixed-strength attention boosting (VHR, ClearSight, SRF) applies the same intervention at every generation step regardless of whether the model is currently grounded or drifting. Over-boosting grounded steps wastes budget; under-boosting risky steps misses the intervention window.

**Solution — Token-level adaptive boosting:**

```
# At each generation step t:
r_{t-1} = VGE from previous step  # hallucination risk estimate
m_t = 1 + (m_vis^max − 1) · r_{t-1}  # adaptive boost magnitude

# Apply:
attn_logits[image_tokens] *= m_t      # boost visual tokens
attn_logits[text_tokens]  *= m_txt^max  # suppress text tokens (fixed)
```

VGE combines predictive entropy H(y) with visual grounding score (cross-modal alignment between image and current hidden state).

**Details:**

- Training-free; applied across all transformer layers at each generation step
- No head selection — all heads boosted uniformly

- Not query-conditioned at the image-token level (all visual tokens boosted equally)
- Tested on modern architectures: Qwen3-VL-8B, InternVL3.5-8B ✓

**Arch:** LLaVA-NeXT-7B, Qwen3-VL-8B, InternVL3.5-8B | **Datasets:** CHAIR (COCO), AMBER, SHR

**Results:**

| Model | CHAIRs ↓ | CHAIRi ↓ |
|---|---|---|
| LLaVA-NeXT-7B baseline | 33.80 | 8.46 |
| + AdaVBoost | 28.80 (−14.8%) | 7.66 |
| Qwen3-VL-8B baseline | 58.80 | 10.57 |
| + AdaVBoost | 46.00 (−21.8%) | 8.38 |

**GAPS / Ideas for us:**

- No head selectivity — boosts all heads including language-biased ones; SRF's top-20% visual heads is more surgical
- No image-token selectivity — VGE scores the *step*, not individual patches; SRF's CLIP saliency selects *which* image tokens to boost per query
- VGE requires computing cross-modal alignment at every step — inference overhead
- Not evaluated on POPE, MMVP, or VLM Bias
- **Complementary idea:** Combine AdaVBoost's adaptive alpha (scale with per-step VGE) with SRF's spatial saliency mask. SRF fixes *where* to boost; AdaVBoost fixes *how much* to boost at each step. Currently SRF uses fixed α=4.0 — VGE-adaptive α is a direct upgrade.

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# PADE

**arXiv:** 2602.15556 | **Date:** Feb 2026 | **Venue:** arXiv
**Authors:** Guangtao Lyu, Qi Liu, Chenghao Xu, Jiexi Yan, Muli Yang, Xueting Li, Fen Fang, Cheng Deng

**Core Intuition**

As information flows through transformer layers, attention to visual tokens isn't constant — it rises and falls. The "positive dynamics" (layers where attention *increases* to a visual token) are the signal; negative dynamics are noise. Aggregate only the increases across layers → you get a saliency map of the most semantically active visual regions. Boost those. Crucially: instead of suppressing instruction tokens (which hurts reasoning), compensate by reducing attention to *system* tokens — those are truly irrelevant.

**Problem:** ClearSight/VAF suppresses system-prompt tokens AND instruction tokens together; suppressing instruction tokens degrades the model's ability to follow complex instructions. Existing inter-layer analysis methods use raw attention values, which are polluted by attention sinks and positional artifacts.

**Solution — Positive Attention Dynamics (PAD) + MAD scaling:**

```
# PAD map — aggregate inter-layer attention increases only:
PAD_v = Σ_l max(0, A_l[→image] − A_{l-1}[→image])

# Per-head adaptive boost via MAD (Median Absolute Deviation):
P̂_{l,h} = MAD(Z_{v,l,h}) × P̃_{l,h}   # scales boost by head's natural variability

# System-token compensation:
Ẑ = Z + α·P̂ − β·Z[→system_tokens]   # boost visual, suppress system (not instruction)
```

Applied in the final transformer layer only.

**Details:**

- Training-free; single forward pass
- Per-head scaling via MAD avoids overshooting heads with naturally large logit variance
- Only the last layer is modified — lighter intervention than ClearSight's layers 8–15

**Arch:** LLaVA-1.5-7B, Qwen-VL | **Datasets:** CHAIR, POPE, AMBER, VizWiz, MME, LLaVA-Wild

**Results (LLaVA-1.5-7B):**

| Method | CHAIRs ↓ | POPE adversarial ↑ | MME ↑ |
|---|---|---|---|
| Baseline | 55.1 | ~76% | ~1800 |
| ClearSight/VAF | -4.2 | — | — |
| **PADE** | **48.6 (-6.5)** | 78.47% | 1892 |

**GAPS / Ideas for us:**

- PAD map uses all heads equally — no vision-aware head selection; SRF targets top-20% visual heads
- Not query-conditioned: same PAD map for all questions on the same image

- Final-layer-only intervention may miss mid-layer cross-modal fusion; SRF uses layers 8–15 (Qwen) where fusion occurs
- **Direct differentiator from SRF:** PADE's saliency = inter-layer attention increases (internal, unsupervised). SRF's saliency = CLIP cross-modal similarity (external, query-conditioned). Different signal sources → different token selections per question.
- **Idea:** Use PAD map as a secondary saliency signal alongside CLIP — average of CLIP sim and PAD could be more robust than either alone.

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## Focus Matters

**arXiv:** 2604.03556 | **Date:** Apr 2026 | **Venue:** arXiv
**Authors:** Sohyeon Kim, Sang Yeon Yoon, Kyeongbo Kong

> **Core Intuition**
>
> The vision encoder (ViT) doesn't process all patches equally at all layers. Attention concentration goes through three phases: *diffuse* (early layers, broad), *focus* (middle layers, concentrated), *re-diffuse* (late layers, spreading again). The focus phase is when the encoder "decides" which patches matter. By suppressing irrelevant patches specifically during the focus phase — using DPP to keep a diverse rather than redundant set — you force the encoder to commit to better visual evidence before it ever reaches the LLM.

**Problem:** Most VLM attention intervention papers modify the LLM decoder's attention. But the visual bottleneck is already created by the vision encoder (ViT) before any cross-modal fusion — noisy patches pollute the image representation that gets fed to the LLM.

**Solution — Phase detection + DPP suppression in ViT:**

```
# Phase detection per layer l:
R_l = E[max attention at l] / E[entropy at l]  # high R = focused phase
Focus phase = arg where R_l peaks (layers 7–11 for LLaVA's ViT)

# DPP token selection within focus phase:
kernel K_ij = exp(−||v_i − v_j||² / σ²)  # based on patch embeddings
select_set S = DPP(K, budget=top-k)       # diverse + high-relevance patches

# Masking in focus phase layers:
attn_logits[token ∉ S] = −∞   # suppress non-selected patches
attn_logits[token ∈ S] = 0    # keep selected (no explicit boost)
```

**Details:**

- Intervenes in the **vision encoder (ViT)**, not the LLM decoder — different intervention point from all other methods in this list
- DPP maintains diversity: avoids selecting 10 patches all from the same corner
- Overhead: +0.212s per sample (ViT encoder only, 3× original encoder time)
- Only tested on LLaVA-1.5; not on Qwen2.5-VL (which uses a different ViT/dynamic-resolution setup)

**Arch:** LLaVA-1.5-7B/13B | **Datasets:** CHAIR, POPE, AMBER

**Results (LLaVA-1.5-7B):**

| Method | CHAIRs ↓ | POPE random ↑ | POPE adv ↑ |
|---|---|---|---|
| Baseline | 45.0 | 87.4 | 79.3 |
| **Focus Matters** | **28.8 (−36%)** | 86.4 | 78.5 |

Note: Large CHAIR gain but slight POPE regression — ViT-level suppression hurts existence judgments.

**GAPS / Ideas for us:**

- **Different intervention point:** ViT encoder vs. SRF's LLM decoder attention. These are complementary and could be stacked.
- POPE regression: suppressing patches in the ViT can remove the very patches needed to answer "Is there a X?" — SRF's LLM-level boost can compensate for this
- Not query-conditioned: DPP selection doesn't know what question will be asked; SRF's CLIP saliency is query-conditioned
- Qwen2.5-VL uses dynamic resolution tiling (448px tiles) — phase detection may differ across tiles, needs investigation
- **Key differentiator:** SRF leaves the visual representation intact, works in the LLM's cross-modal fusion layers. Focus Matters modifies the visual representation before LLM fusion. Different trade-offs: SRF preserves visual fidelity, Focus Matters improves input quality.

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# CAST

**arXiv:** 2605.04641 | **Date:** Jun 2026 | **Venue:** arXiv
**Authors:** Qiming Li, Zekai Ye, Xiaocheng Feng, Weihong Zhong, Libo Qin, Ruihan Chen, Lei Huang, Baohang Li, Kui Jiang, Yaowei Wang, Ting Liu, Bing Qin

**Core Intuition**

When you describe an image (caption it), your attention heads naturally align with visual content more strongly than when answering abstract questions. These "caption-sensitive" heads are the model's natural visual grounding machinery. CAST finds them via a quick SVM classifier on head outputs, precomputes shift vectors (direction = "caption mode" minus "non-caption mode"), then injects those vectors at inference for any query type. It's like giving the model permission to use its visual grounding heads even when not captioning.

**Problem:** Attention heads specialized for visual grounding are underutilized during VQA/hallucination-prone tasks compared to captioning tasks. No existing method leverages caption-task behavior as a supervision signal for attention steering.

**Solution:**

```
# Offline: identify top-K caption-sensitive heads (K=100):
Train SVM on head output features: caption vs. non-caption queries
Select top-100 heads by SVM importance score

# Precompute shift vectors S(l,h) per selected head:
S(l,h) = mean_over_batch [O_{caption}(l,h) − O_{non-caption}(l,h)]

# At inference — inject for all queries:
H^{l+1} = H^l + Σ_{(l,h) ∈ top-K} [O(l,h) + I(l,h)·α·S(l,h)] · W_o^l
# α=1.5; I(l,h) = indicator for selected heads
```

**Details:**

- Training-free at inference (offline calibration step needed to compute S)
- Calibration: needs captioning-annotated queries to compute shift vectors
- 1.0× inference overhead (precomputed S injected as constant vector addition)
- Tested on 5 architectures ✓

**Arch:** LLaVA-1.5-7B (primary), 4 additional models | **Datasets:** POPE (all 3 splits), CHAIR, MMHal-Bench, MMVet

**Results (LLaVA-1.5-7B):**

| Split | Baseline | CAST |
|---|---|---|
| POPE random | 83.29% | **89.87%** (+6.58%) |
| POPE popular | 81.88% | **88.32%** (+6.44%) |

| POPE adversarial | 78.96% | **84.27%** (+5.31%) |
| CHAIRs | — | −6.43 pts avg |

**GAPS / Ideas for us:**

- Calibration requires caption-labeled queries — not zero-shot portable across arbitrary new VLMs
- Shift vectors S are precomputed once on a fixed calibration set; not adaptive per sample or query type
- Head selection by caption sensitivity is indirect: not the same as selecting vision-aware heads by attention magnitude (VHR/SRF)
- **Key differentiator:** CAST steers *what the model outputs* from visual heads (output space); SRF boosts *which image tokens get attended to* (attention space). CAST's shift is a fixed direction; SRF's boost is query+image conditioned.
- **Inspiration:** The idea of identifying heads by their behavior on a reference task (captioning) rather than by attention magnitude is novel. Could apply to SRF: identify heads by caption-task VHD divergence, not just mean image attention.
- Not tested on Qwen2.5-VL, MMVP, or VLM Bias.

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## ILVAD

**arXiv:** 2605.20965 | **Date:** Jun 2026 | **Venue:** arXiv
**Authors:** Yutong Xie, Zhenglin Hua, Ran Wang, Wing W. Y. Ng, Xizhao Wang, Yuheng Jia

> **Core Intuition**
>
> A single layer's attention map is noisy — dominated by attention sinks and positional biases. But if a visual token is genuinely important, it should attract consistent attention *across many layers*, not just spike in one. By computing layer-by-layer attention differences (discrepancy), you filter out layer-specific noise and sinks, leaving only tokens with stable cross-layer importance. These are the "correct visual evidence" tokens. Then boost them hard (exponentially) and additionally tie the text tokens most connected to them.

**Problem:** Single-layer saliency is unreliable due to attention sinks, positional artifacts, and layer-specific noise. Multi-layer aggregation methods (rollout, GradCAM) sum all layers equally, diluting the signal with noise layers.

**Solution — Inter-layer discrepancy saliency + exponential boost:**

```
# Step 1: Build inter-layer discrepancy saliency map:
saliency = Σ_l |A_l[first_T_tokens → image] − A_{l-1}[first_T_tokens →
image]|
# T=10 initial tokens; subtraction cancels persistent sinks

# Step 2: Visual evidence amplification:
attn_logits[evidence_tokens] *= exp(α · attn_weight)   # α=3–5

# Step 3: Text grounding — boost text tokens connected to visual evidence:
text_score = correlation(text_attn, visual_evidence_attn)
attn[text_tokens with high score] *= β_scale   # β=0.2–0.5
```

**Details:**

- Training-free; single forward pass with hook on attention matrices
- Uses first T=10 token positions for saliency — computationally cheap
- Exponential scaling (vs linear in VAF/SRF) creates sharper contrast between evidence and noise
- Tested on 5 architectures

**Arch:** LLaVA-1.5-7B (primary), 4 additional models | **Datasets:** CHAIR, POPE, MMHal-Bench, MME, CapMAS

**Results (LLaVA-1.5-7B):**

| Method | CHAIRs ↓ | POPE acc ↑ | MMHal ↑ |
|---|---|---|---|
| Baseline | 48.6 | 84.50% | 2.01 |
| VHR | ~45 | — | — |
| **ILVAD** | **32.6 (−32.9%)** | **85.76%** (+1.26%) | **2.22** (+10.4%) |

**GAPS / Ideas for us:**

- Saliency from first-T token attention — initial tokens tend to look at the most salient regions globally; not query-conditioned on the specific question being asked
- Exponential scaling is more aggressive than linear boost; can destabilize attention distribution if over-applied
- Text grounding step (β) adds a second hyperparameter specific to task type
- **Direct differentiator from SRF:** ILVAD's saliency = inter-layer internal attention consistency (unsupervised, global). SRF's saliency = CLIP text-image similarity (external, query-conditioned). ILVAD doesn't know what's being asked; SRF selects different tokens for "Is there a chair?" vs. "Is there a clock?" on the same image.

- **Inspiration:** The inter-layer discrepancy idea is a cheap way to filter attention sinks without needing external CLIP. Could serve as a fallback saliency signal when CLIP similarity is low (our `clip_fallback_thresh` path) — use ILVAD-style internal saliency instead of uniform boost.

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## Synthesis & Research Directions

| Gap | Promising Approach |
|---|---|
| VHR uses fixed α=2 | Adaptive per-layer scale; calibrate on VLM Bias samples |
| ClearSight hardcoded layers 8–15 for LLaVA | Empirically find Qwen2.5-VL fusion layers via saliency (S_vt metric) |
| Contrastive methods (VCD/ICD) hurt coherence | VAF-style attention boost as drop-in replacement — no quality degradation |
| Visual attention sinks waste capacity | VAR sink redistribution + VHR head selection = combined method |
| VLM Bias: model ignores image in favor of language prior | Localization heads to *diagnose* where model looks; BASS to suppress background |
| OPERA only helps long-form generation | Skip for MCQ/yes-no; useful if we add captioning eval |
| All methods tested on LLaVA, not Qwen2.5-VL | **Our experiments fill this gap** — first systematic study on Qwen2.5-VL |
| No method targets language-prior bias directly | VAF suppression mask on system-prompt tokens is the closest attempt |
| AIR selects tokens by visual prototype distance — query-agnostic | SRF-CLIP/HSSA is explicitly query-conditioned: different question → different tokens boosted on same image. This is the novel axis AIR doesn't cover. |
| Saliency-R1 achieves faithful visual grounding but requires LoRA+GRPO+annotated bounding boxes | SRF is the training-free equivalent. Saliency-R1's logit decomposition (training-free component) is a useful diagnostic for before/after SRF visualizations. Future: use SRF saliency mask as GRPO reward signal, removing annotation dependency. |
| SPIN suppresses inattentive heads but doesn't select which image tokens to boost | SRF adds query-conditioned image-token selectivity on top of head selection — different token set per question on the same image |

| Gap | Promising Approach |
|---|---|
| FlashVLM does query-conditioned token selection but as pruning (efficiency), not attention boosting (grounding) | SRF's CLIP saliency is spatial and externally grounded; FlashVLM's LLM-space sim is cheaper but less interpretable. Could use LLM-space sim as saliency fallback when CLIP is unavailable. |
| CAAC applies uniform boost to all image tokens, no query signal | Combine CAAC's confidence gate (higher α when model is uncertain) with SRF's saliency mask — adaptive alpha per sample |
| SRF-E zeros full image for null pass → suppresses format tokens in VLM Bias | Use spatial CFG instead: zero only CLIP-salient region patches → null pass retains format context, contrastive signal isolates the relevant region's contribution |
| No method uses Δ_content (visual grounding score) as an adaptive signal | Use PPL_Real − PPL_Noise per sample as a dynamic alpha scaler for SRF: weakly-grounded samples get stronger boost |
| SRF uses fixed α=4.0 regardless of per-step hallucination risk | AdaVBoost shows VGE-adaptive α at each step. Combine: SRF's *where* (CLIP saliency mask) + AdaVBoost's *how much* (VGE risk per step) = best of both |
| PADE/ILVAD use internal attention patterns as saliency — no CLIP | Use inter-layer discrepancy or PAD map as `clip_fallback` saliency when CLIP max_sim is low; avoids uniform boost on hard queries |
| CAST uses caption-task head sensitivity, not attention magnitude, for head selection | Alternative head selection criterion: heads that diverge most on caption vs. VQA tasks. More semantically motivated than mean image attention score. |
| Focus Matters operates in ViT encoder (pre-LLM), not decoder | Complementary intervention: Focus Matters cleans the visual representation entering the LLM; SRF guides how that representation is used. Could stack both — no architectural conflict. |
| ILVAD's exponential scaling (exp(α·attn)) is sharper than SRF's additive logit | Experiment: replace SRF's additive boost with multiplicative exp-scaling for harder cases (VLM Bias). Risk: destabilization at large α. |